

1.2 Classwork: Speed, Velocity, and Unit Conversion

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ mi} = 5280 \text{ ft} \quad 1 \text{ mi} \approx 1.61 \text{ km} \quad 1 \text{ m} \approx 3.28 \text{ ft} \quad 1 \text{ ft} = 12 \text{ in}$$

$$1 \text{ km} = 1000 \text{ m} \quad 1 \text{ in} = 2.54 \text{ cm} \quad 1 \text{ h} = 3600 \text{ s} \quad 1 \text{ min} = 60 \text{ s}$$

1. A student walks 24.0 meters down the hallway in 18.5 seconds.
 - (a) Find the average speed in meters per second, rounded to three significant figures.
 - (b) How many significant figures does your answer have, and why not more?

2. Convert each speed. Write each conversion as a fraction and show the units cancelling.
 - (a) A bus on Via del Corso travels at 30.0 kilometers per hour. What is this speed in meters per second?
 - (b) A train travels at 25.0 meters per second. What is this speed in kilometers per hour?

3. A fastball leaves the pitcher's hand at 95 miles per hour.
 - (a) What is this speed in feet per second?
 - (b) What is this speed in meters per second?

5. The speed of sound in air is 343 meters per second. You see a lightning flash and hear the thunder 4.0 seconds later. How far away was the lightning, in meters? In miles?

6. Our building is a New York brownstone, and the classroom is on the 6th floor — so from the entrance you climb 5 floors. Each floor is one flight of 18 steps, and each step is 8.0 inches high.
- (a) How many steps do you climb in total?
 - (b) What is the total elevation gain, in feet?
 - (c) Climbing steadily at 1.2 steps per second, how long does the climb take? Give your answer in seconds, and again in minutes and seconds.
 - (d) What is your average vertical speed, in feet per minute?
7. Challenge: a car's speedometer reads 30 miles per hour. A student converts this to kilometers per hour and writes 48.28032 km/h. Explain what is wrong with this answer, and write a better one.